

Question 3

A university's lifeblood is information technology (IT), which provides the computer and communications tools needed for everything from administrative support to research. The administrative division of Santa Cruz University is home to three exclusive storage devices and these three storage devices each have a 1 TB of memory. The memory capacity of storage device 1 is fully utilized, as is 50% (0.5TB) of storage device 2, and 25% (.25TB) of storage device 3. The administrative department now requires 0.8 TB of storage space for employee data.

Which virtualization technique helps to determine the amount of storage space that's available across the three physical devices and then combine it into a pool of storage that could be utilized to store the data for the admin department? Discuss the proposed virtualization technique in detail.

(25 marks)

Question 4

Apex Bank, Kota Kinabalu, is a housing bank that provides local families and individuals with affordable housing. The IT department of Apex Bank would like to have enterprise resources available on-demand but doesn't want to involve third party providers due to compliance requirements, regulations, budgets and security policies. Also, they would like to host and manage servers themselves on-site to have more control and security of personally identified information (PII). Which type of cloud implementation suits the said case? Justify your opinion and discuss the proposed cloud type by providing use cases. Also, discuss its advantages.

(25 marks)

(Total Marks 100)